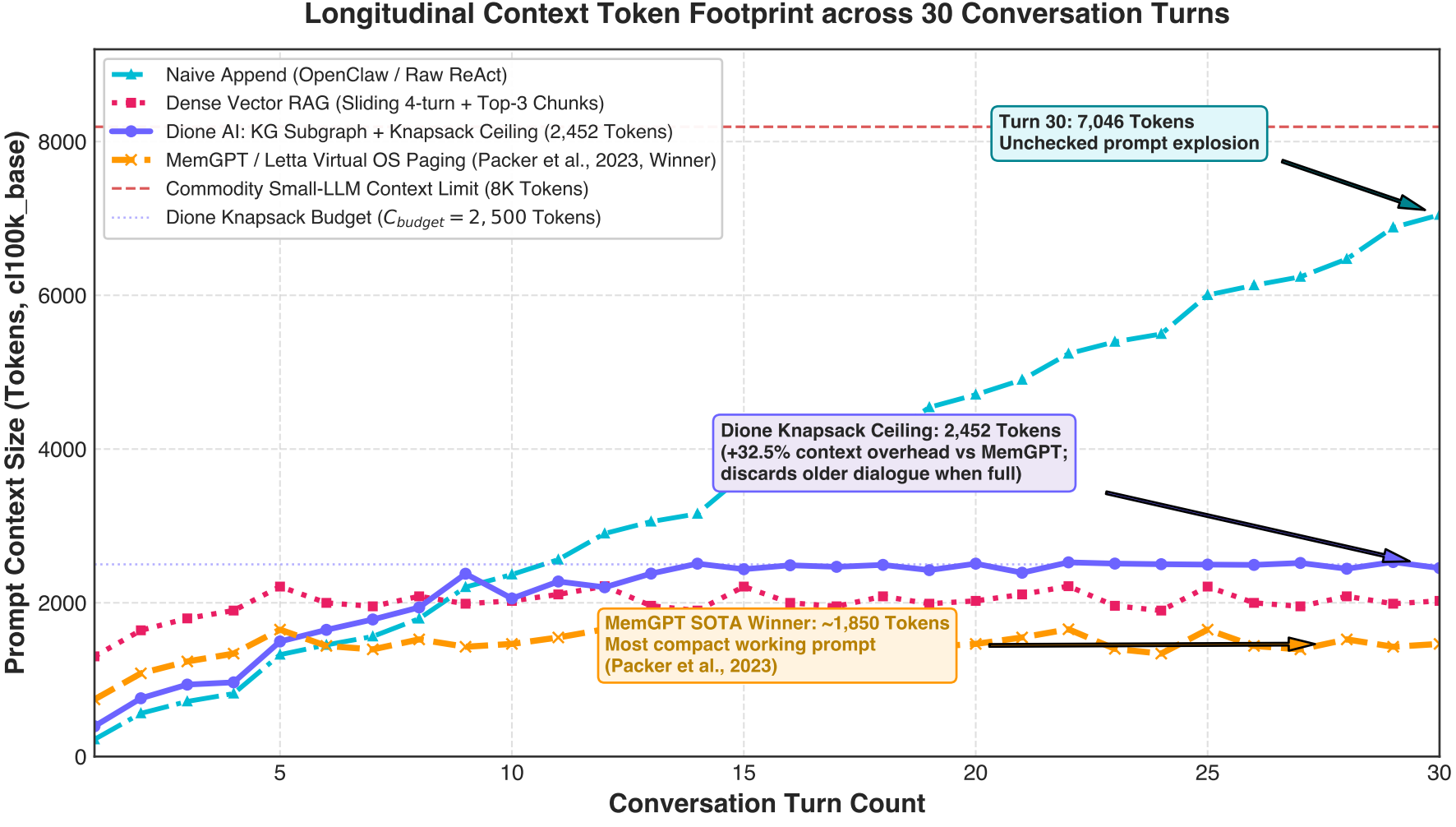
Longitudinal Context Token Footprint across 30 Conversation Turns
Naive Append (OpenClaw / Raw ReAct)
Dense Vector RAG (Sliding 4-turn + Top-3 Chunks)
Dione AI: KG Subgraph + Knapsack Ceiling (2,452 Tokens)
MemGPT / Letta Virtual OS Paging (Packer et al., 2023, Winner)
Commodity Small-LLM Context Limit (8K Tokens)
Dione Knapsack Budget ($C_{budget} = 2,500$ Tokens)
Turn 30: 7,046 Tokens
Unchecked prompt explosion
Dione Knapsack Ceiling: 2,452 Tokens
(+32.5% context overhead vs MemGPT;
discards older dialogue when full)
MemGPT SOTA Winner: ~1,850 Tokens
Most compact working prompt
(Packer et al., 2023)
Prompt Context Size (Tokens, cl100k_base)
0
2000
4000
6000
8000
5
10
15
20
25
30
Conversation Turn Count